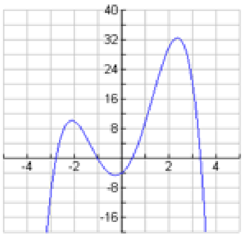
1.2: Characteristics of Polynomial Functions

Example 1:
Match each of the following equations with its graph. Fill in the blanks below.

$f(x) = 2x^3 - 4x^2 + x + 1$	$g(x) = -x^4 + 10x^2 + 5x - 4$	$h(x) = -2x^5 + 5x^3 - x$	$p(x) = x^6 - 16x^2 + 3$

a)



Equation: ____________________

\# of x-intercepts: ____________________

\# absolute max: ____________________

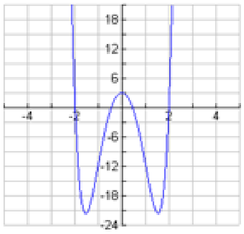
\# absolute min: ____________________

\# local max: ____________________

\# local min: ____________________

total # locals: ____________________

b)



Equation: ____________________

\# of x-intercepts: ____________________

\# absolute max: ____________________

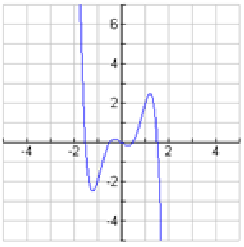
\# absolute min: ____________________

\# local max: ____________________

\# local min: ____________________

total # locals: ____________________

c)



Equation: ____________________

x-intercept: ____________________

\# absolute max: ____________________

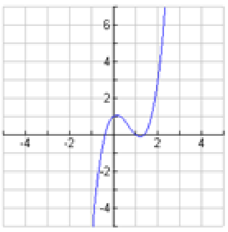
\# absolute min: ____________________

\# local max: ____________________

\# local min: ____________________

total # locals: ____________________

d)



Equation: ____________________

x-intercept: ____________________

\# absolute max: ____________________

\# absolute min: ____________________

\# local max: ____________________

\# local min: ____________________

total # locals: ____________________

Example 2:

The following table of values represents a polynomial function. Use finite differences to determine the degree of the polynomial function, the sign of the leading coefficient, and the value of the leading coefficient.

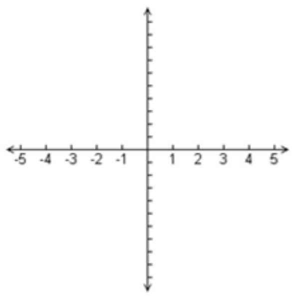
x	y	First Differences	Second Differences	Third Differences	Fourth Differences
-3	-54				
-2	-16				
-1	-2				
0	0				
1	2				
2	16				
3	54				
4	128				

Example 3:

Consider the function $f(x) = -x^3 + 2x^2 + 3x$.

a) How do the degree and the sign of the leading coefficient correspond to the end behaviour of the polynomial function?

b) Sketch a graph of the polynomial function. Include all intercepts.



c) What can you tell about the value of the third differences for this function? Calculate.

HOMEWORK: p. 26 # 7, 8, 9a, 11, 12, 15, 17